Suei-Wen Chen

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EDUCATION

ETH Zürich *Master in Data Science* Sep 2024 – Present

Interest: Theoretical Foundations / Reinforcement Learning / LLM Agents, Weighted GPA 5.80/6.00

New York University Abu Dhabi *Bachelor of Science* Aug 2019 – May 2023

Major: Mathematics, Minor: Computer Science, Cumulative GPA: 4.0/4.0 (summa cum laude)

PROFESSIONAL EXPERIENCE

Teaching Assistant | *Department of Computer Science, ETH Zürich* Aug 2025 – Present

Co-lead the design of coding projects/exercises and give lectures in the following courses:

- From Data to Decisions (Summer 2026): Information Theory, Explainable AI, Security, RL, Vector DBs and RAGs, Agents
- AI in Industry (Fall 2025, Spring 2026, Fall 2026): LLMs, Agents, Recommender Systems, RL

Research Assistant | *Department of Mathematics, New York University Abu Dhabi* Sep 2023 – Aug 2024

- Supervised undergraduate student theses
- Investigated foundations of RL, language models, differential privacy and bias of multi-modal data classification

Data Science Intern | *Swftbox Delivery Service LLC, UAE* Mar 2022 – Aug 2023

- Optimized last-mile logistics with data-driven solutions
- Implemented geocoding pipeline, prediction algorithms, optimizations for operations, and vehicle routers

RESEARCH

Mathematical Capabilities of LLMs | *Center for Interdisciplinary Data Science and AI, NYUAD* Sep 2025 – Sep 2026

Design agentic workflow and benchmark for LLM-assisted research in mathematics

Finite-Sample Analysis of the Monte Carlo Exploring Starts Algorithm for Reinforcement Learning | Jan 2024-Aug 2024

Established convergence rate for Monte Carlo algorithm in episodic reinforcement learning (arXiv:2410.02994)

How Bad is Training on Synthetic Data? A Statistical Analysis of Language Model Collapse | Nov 2023 – Apr 2024

Published in *Conference on Language Modeling (COLM)* 2024, arXiv:2404.05090

Quantified how training language models on generated data reduces output diversity (model collapse)

Markov Chains, Mixing Times and Functional Inequalities | *Undergraduate Thesis* Mar 2022 – May 2023

Studied the mixing properties of Markov chains in relation to functional inequalities based on decomposition (arXiv:2410.17301)

PROJECTS

Reinforcement learning on vehicle routing problems | *D-INFK, ETH Zürich* May 2026 – Jun 2026

Policy-gradient method with attention-based feature encoding for capacitated vehicle routing problem on road networks

Tool-Gym: Tool-Calling Data Generation Pipeline | *Data Science Lab, ETH AI Center* Sep 2025 – Dec 2025

Constructed RLVR data pipeline for tool calling and conducted LLM post-training

Smart addressing system | *Swftbox Delivery Service LLC* Apr 2022 – Aug 2022

Addresses recognition pipeline with NLP and graph methods; outperformed commercially available services

COMPETITIONS, HONORS & AWARDS

Datathon 2025 | *ML/AI Hachathon, Analytics Club at ETH, Zürich* Apr 2025

Achieved competitive performance of web-search LLM reasoning agent on multi-hop QA dataset

Nominated for Wexler PhD Fellowship | *Faculty of Computing and Data Science, Boston University* Mar 2024

Full Scholarship for Undergraduate Studies | *New York University Abu Dhabi* Sep 2019 – May 2023

LANGUAGES AND SKILLS

Programming Languages/Skills |

- **Programming**: Git, Python (PyTorch, Scikit-learn, Pandas, NumPy, SciPy), Slurm, Docker, data visualization
- **LLMs/Agents**: MCP, orchestration (LangGraph), RAGs, PEFT, post training, benchmarks, security
- **Machine Learning**: statistical learning, computer vision, NLP, RL, explainability
- **Databases**: PostgreSQL, Spark, Neo4j, MongoDB, vector databases, hybrid search
- **Analysis**: optimization & operations research (Google OR-tools), probability theory, statistical analysis

Languages |

Chinese (Traditional) (native), **English** (C2), **German** (A2), **Arabic** (A2)

VOLUNTEER AND LEADERSHIP EXPERIENCE

Vice Director | *Global Devotion Association, Taiwan* Jun 2017 – Aug 2023

Organized cultural exchange programs between Taiwan and the US, summer camps, volunteering and community outreach

REFERENCES

References available upon request